数学笔记

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目录

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Proposition 23.1. *If the manifold M has r connected components, then its de Rham cohomology in degree 0 is $H^0(M) = \mathbb{R}^r$.*

Proof. Since there are no exact 0-forms other than 0,

$$H^0(M) = Z^0(M) = \{\text{closed 0-forms}\}.$$

Supposed f is a closed 0-form on M, i.e., f is a C^∞ function on M such that $df = 0$. On any chart $(U, x^1, \ldots, x^n)$,

$$df = \sum \frac{\partial f}{\partial x^i}\, dx^i.$$

Thus, $df = 0$ on U if and only if all the partial derivatives $\partial f/\partial x^i$ vanish identically on U. This in turn is equivalent to f being locally constant on U. Hence, the closed 0-forms on M are precisely the locally constant functions on M. Such a function must be constant on each connected component of M. If M has r connected components, then a locally constant function on M is simply an ordered set of r real numbers. Thus, $Z^0(M) = \mathbb{R}^r$. □

Proposition 23.2. *On a manifold M of dimension n, the de Rham cohomology $H^k(M) = 0$ for $k > n$.*

Proof. At any point $p \in M$, the tangent space T_pM is a vector space of dimension n. If ω is a k-form on M, then $\omega_p \in A_k(T_pM)$, the space of alternating k-linear functions on T_pM. By Corollary 3.31, if $k > n$, then $A_k(T_pM) = 0$. Hence, for $k > n$, the only k-form on M is the zero form. □

The de Rham cohomology $H^*(M)$ is an anticommutative graded ring, with wedge as the multiplication; anticommutative means :

$$a \cdot b = (-1)^{k\ell} b \cdot a$$

where $a \in A^k, b \in A^\ell$ for the graded ring $A = \oplus_{k=0}^{\infty} A^k$; in graded ring the ring multiplication sends $A^k \times A^\ell$ to $A^{k\times\ell}$.

Suppose $F : N \to M$ is a C^∞ mapping of manifolds; because $F^*(\omega \wedge \tau) = F^*\omega \wedge F^*\tau$ for differential forms ω and τ on M, the pullback map $F^* : H^*(M) \to H^*(N)$ is a ring homomorphism. If F is diffeomorphism, then F^* is isomorphism.

To sum up, de Rham cohomology gives a contravariant functor from the category of C^∞ manifolds to the category of anticommutative graded rings. In this way the de Rham cohomology becomes a powerful diffeomorphism invariant of C^∞ manifolds.

The vector space $\Omega^*(M)$ of differential forms on a manifold M together with the exterior d is a cochain complex, the *de Rham complex* of M:

$$0 \to \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \xrightarrow{d} \cdots, \quad d \circ d = 0.$$

It turns out that many of the results on the de Rham cohomology of a manifold depend not on the topological properties of the manifold, but on the algebraic properties of the de Rham complex. To better understand de Rham cohomology, it is useful to isolate these algebraic properties. In this chapter we investigate the properties of a cochain complex that constitute the beginning of a subject known as *homological algebra*.